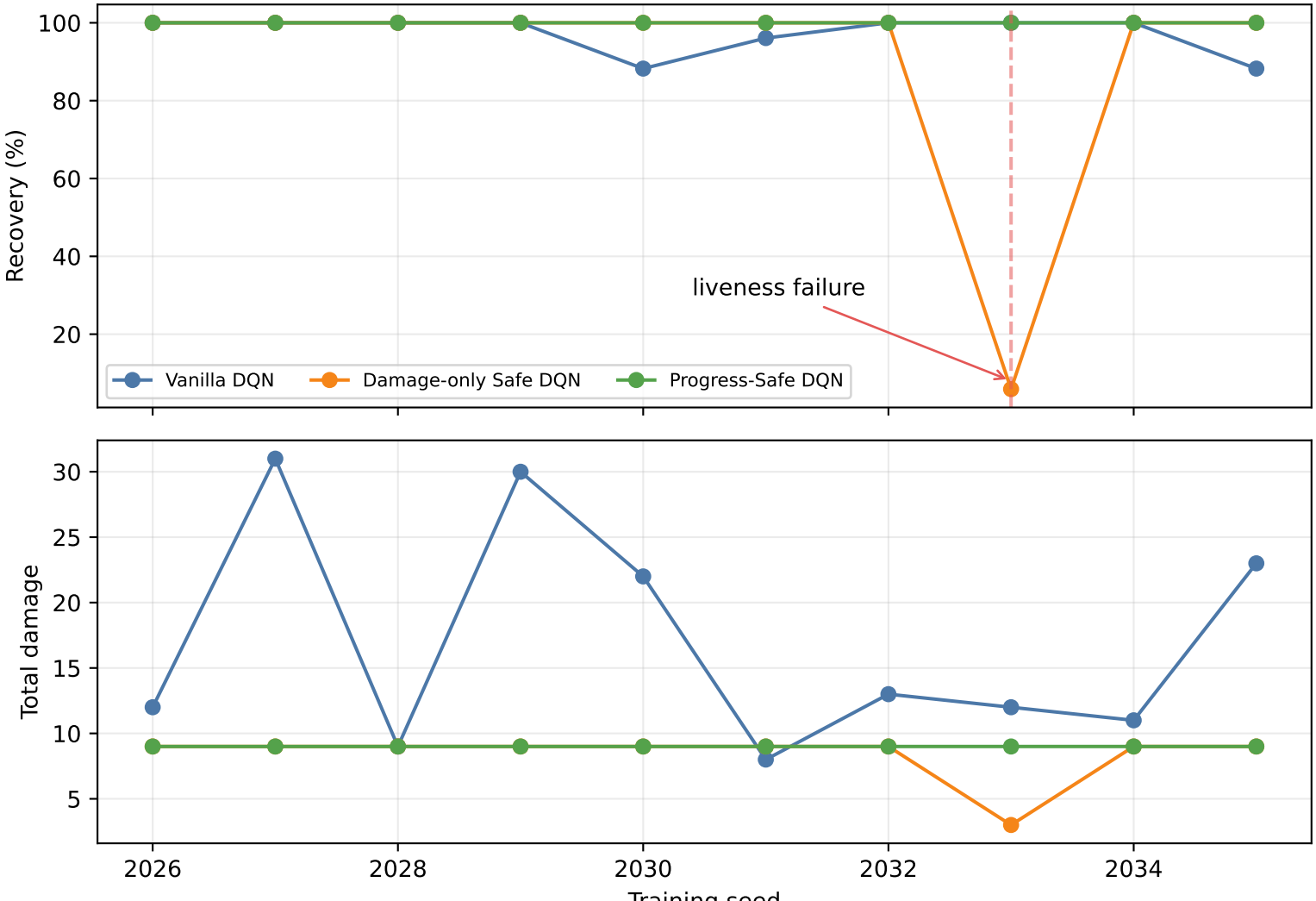
Recovery (%)
100
80
60
40
20
liveness failure
Vanilla DQN
Damage-only Safe DQN
Progress-Safe DQN
Total damage
30
25
20
15
10
5
2026
2028
2030
2032
2034
Training seed